



Computer organization and assembly language
ENCS336

Course Project

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Section : 1

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I. Abstract

This project describes an ID data management system to store student's ids and perform some operations on it.

This helps understand and develop a data analysis program using the assembly language 8086 and emulator Emu8086.

Also, the project report provides the main objectives of the project, data collection along with the specification of the code, the result, and the conclusion that offers insights into the theoretical and simulation aspects of Assembly language programming.

II. Introduction

The design of this project is addressed to create a data management system for the IDs using the assembly language 8086 and the Emu8086 emulator.

The main objective of this program is to create a user friendly interface to perform various operations on the entered ids.

Requirements:

1. Define\declare the necessary data and variables.
2. Data collection
3. Menu
4. Comments

Specifications:

Data declaration: to declare important variables such as the array at the beginning of the code.

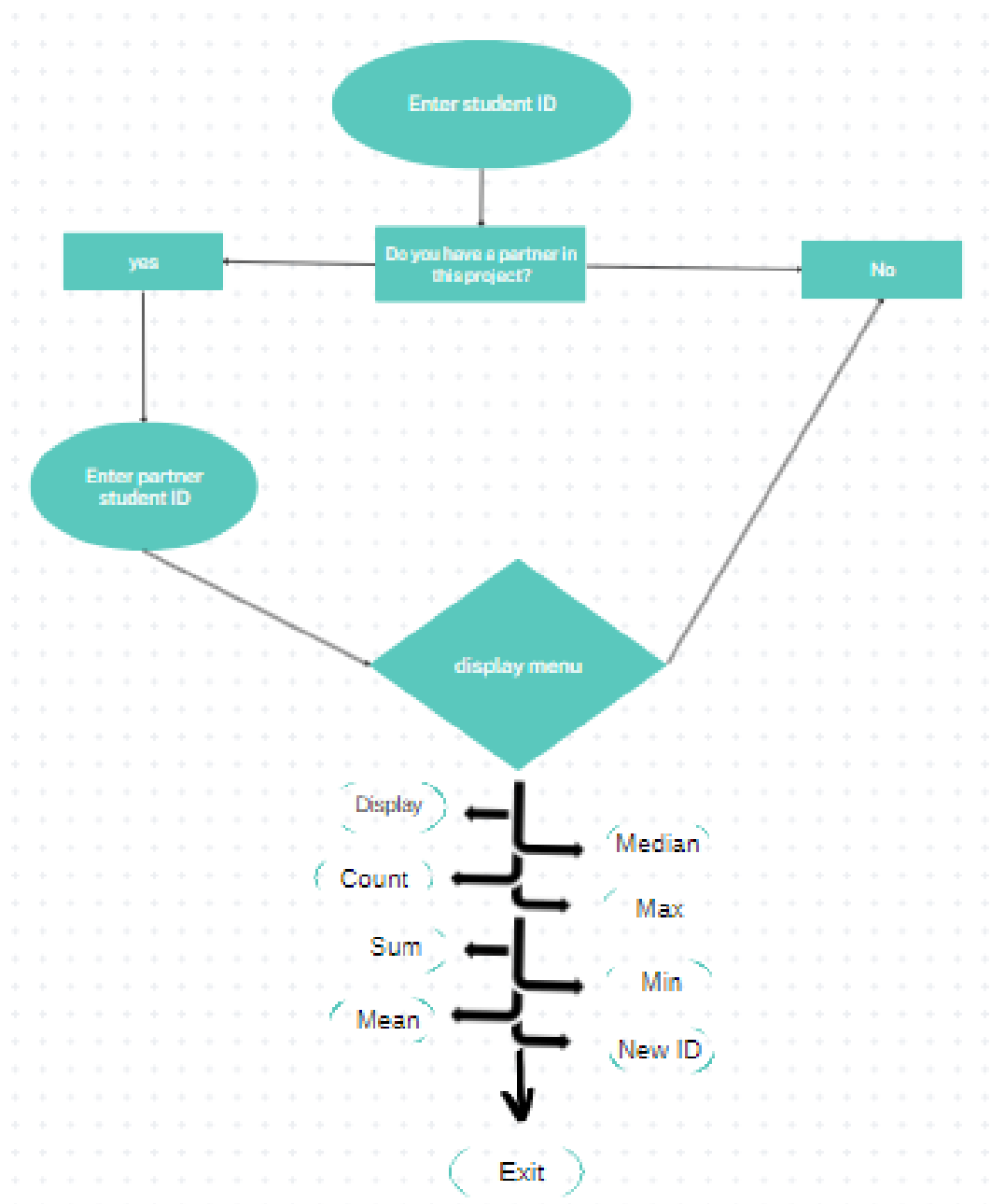
Data collection: use proper inputs and appropriate instructions to collect the ids of the student and their partner and store it in the declared array.

Menu: to implement a menu that has some options that perform some operations on the ids.

By doing these requirements the program should work fine and provides a comprehensive solution for data collecting, managing, and processing.

Comments: add comments to describe code lines.

III. Architecture



IV. Component design

In this system we used various segments, each segments is responsible for different functionalities.

1. Data collection segment:

First for our program to work we need to collect the data which are based on the student's IDs, we let the user enter the student ID and store it in an array of IDs, and checks if there is another partner is participating in this project also and prompts the user to enter the other student ID.

We used prompt to print some sentences like "Enter your ID:"

2. Menu segment:

To display the menu when entering the student's ids which has multiple functions.

The user is going to choose the options from the menu to operate the operations on the array elements and display the results.

Display the menu instruction is to display the menu on the console when running the code after entering the ids.

Then choose from the menu instruction is to determine action to be taken on the array elements.

Handling each menu option by implementing the functions body for each option on the menu.

3. Menu options segments to perform on the array:

As we described in the previous segment we have a menu that has multiple options to be performed on the array, these options are:

Display: to display the array values, which the values are stored like that [1,2,1,1,1,5,3].

Count: to count how many values in the array such as when entering a new ID the count of the array is incremented.

Sum: using a loop, the loop is going to loop around the array elements and add it to the sum later on to be used to find the mean.

Mean: is to calculate the average using the sum and count by calling each method and dividing the sum/count

Call sum

Mov ax, sum

Call count

Mov bx , count

Div ax, bx

Median: to calculate the median of the array elements , the middle data point in the array, first you must take the sum of two elements and then divide by 2.

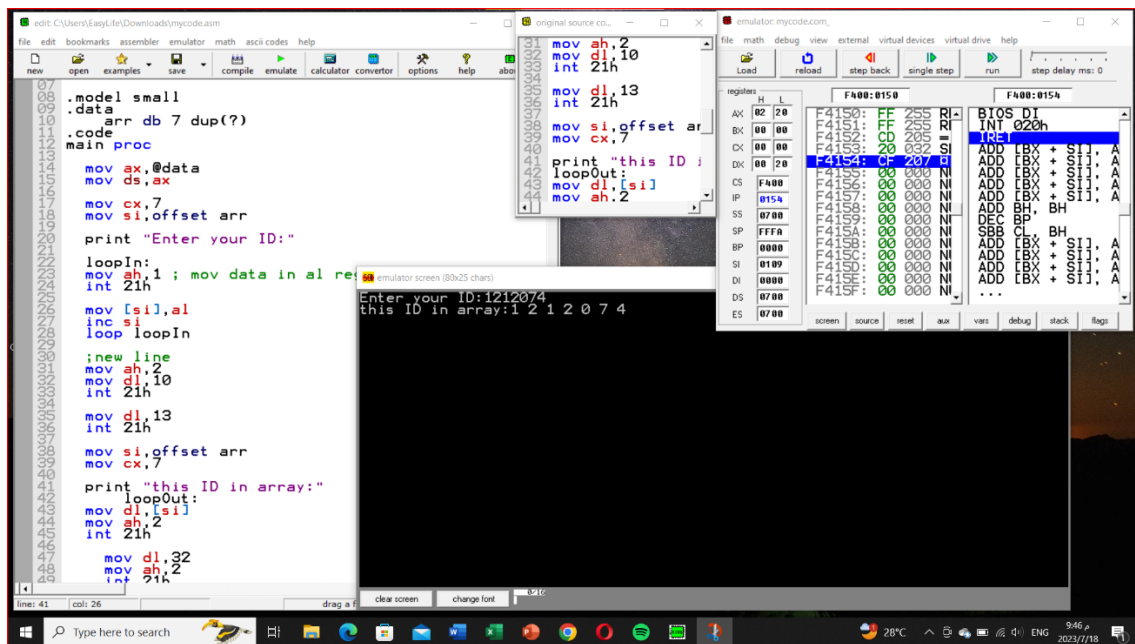
Maximum: to find the maximum number entered in the array by looping around the array elements such as [1,2,1,1,1,5,3,1,2,1,2,0,7,4] the maximum number is 7.

Minimum: to find the minimum number entered in the array like the previous example the minimum number is 0.

New ID: is to enter a new ID in the array.

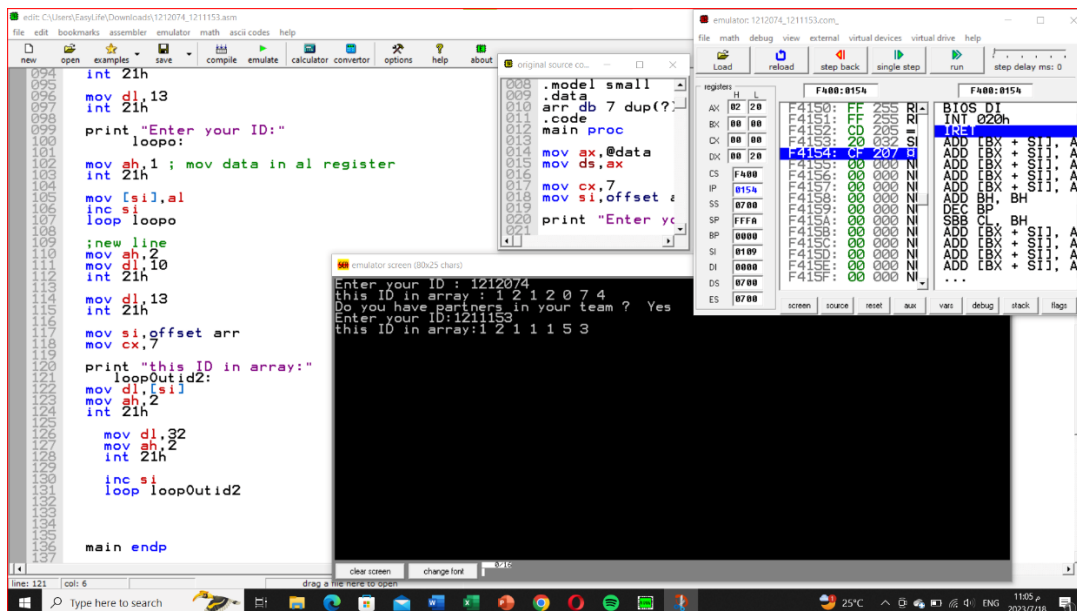
Exit: to exit the menu by choosing the exit option from the menu, 'Exit' we use this word to exit the program.

V. Results and discussion



The previous picture shows the first thing that should be implemented in the code which is the Enter student id and store it in the array.

Then its going to ask if there is a partner in this project which is yes



Then the menu is displayed after entering the second student ID:

```

145 mov ah, 09h ; DOS function to print a string
146 mov dx, offset MenuText
147 int 21h
148
149 Loop:
150 ; Read the user's choice
151 mov ah, 01h ; DOS function to read a character
152 int 21h
153 sub al, '0' ; Convert the character to its numeric value
154
155 ; Handle the user's choice
156 cmp al, 1 ; Check if choice == 1
157 je DisplayOption
158 cmp al, 2 ; Check if choice == 2
159 je CountOption
160 cmp al, 3 ; Check if choice == 3
161 je SumOption
162 cmp al, 4 ; Check if choice == 4
163 je MeanOption
164 cmp al, 5 ; Check if choice == 5
165 je MedianOption
166 cmp al, 6 ; Check if choice == 6
167 je MaxOption
168 cmp al, 7 ; Check if choice == 7
169 je MinOption
170 cmp al, 8 ; Check if choice == 8
171 je NewIdOption
172 cmp al, 9 ; Check if choice == 9
173 je ExitOption
174 jmp InvalidOption ; Jump to invalid input
175
176 ; new line
177 mov ah, 10
178 int 21h
179
180 mov dl, 13
181 int 21h
182
183 DisplayOption:
184 ; Call the Display function "option"
185 push offset IDArray
186
187 ; new line
188 mov ah, 10
189 int 21h

```

original source code:

```

141 int 21h
142
143 mov dl, 13
144 int 21h
145
146 mov ah, 09h ; Print the menu options
147 mov dx, offset MenuText
148 int 21h
149
150 Loop:
151 ; Read the user's choice
152 mov ah, 01h ; DOS function to read a character
153 int 21h
154 sub al, '0' ; Convert the character to its numeric value
155
156 ; Handle the user's choice
157 cmp al, 1 ; Check if choice == 1
158 je DisplayOption
159 cmp al, 2 ; Check if choice == 2
160 je CountOption
161 cmp al, 3 ; Check if choice == 3
162 je SumOption
163 cmp al, 4 ; Check if choice == 4
164 je MeanOption
165 cmp al, 5 ; Check if choice == 5
166 je MedianOption
167 cmp al, 6 ; Check if choice == 6
168 je MaxOption
169 cmp al, 7 ; Check if choice == 7
170 je MinOption
171 cmp al, 8 ; Check if choice == 8
172 je NewIdOption
173 cmp al, 9 ; Check if choice == 9
174 je ExitOption
175 jmp InvalidOption ; Jump to invalid input
176
177 ; new line
178 mov ah, 10
179 int 21h
180
181 mov dl, 13
182 int 21h

```

emulator screen (80x25 chars):

```

Enter your ID : 1211153
this ID in array : 1 2 1 1 1 5 3
Do you have partners in your team ? Yes
Enter your ID:1212074
this ID in array:1 2 1 2 0 7 4
MENU
1. DISPLAY
2. COUNTS
3. SUM
4. MEAN
5. MEDIAN
6. MAX
7. MIN
8. NEW ID
9. EXIT
Menu option, choose wisely ...

```

Menu method body:

```

1396 Menu: ; Menu have many items
1397 ; new line
1398 mov ah, 10
1399 int 21h
1400
1401 mov dl, 13
1402 int 21h
1403 ; Print the menu options
1404 mov ah, 09h ; DOS function to print a string
1405 mov dx, offset MenuText
1406 int 21h
1407
1408 Loop:
1409 ; Read the user's choice
1410 mov ah, 01h ; DOS function to read a character
1411 int 21h
1412 sub al, '0' ; Convert the character to its numeric value
1413
1414 ; Handle the user's choice
1415 cmp al, 1 ; Check if choice == 1
1416 je DisplayOption
1417 cmp al, 2 ; Check if choice == 2
1418 je CountOption
1419 cmp al, 3 ; Check if choice == 3
1420 je SumOption
1421 cmp al, 4 ; Check if choice == 4
1422 je MeanOption
1423 cmp al, 5 ; Check if choice == 5
1424 je MedianOption
1425 cmp al, 6 ; Check if choice == 6
1426 je MaxOption
1427 cmp al, 7 ; Check if choice == 7
1428 je MinOption
1429 cmp al, 8 ; Check if choice == 8
1430 je NewIdOption
1431 cmp al, 9 ; Check if choice == 9
1432 je ExitOption
1433 jmp InvalidOption ; Jump to invalid input
1434
1435 ; new line
1436 mov ah, 10
1437 int 21h

```

emulator screen (80x25 chars):

```

Enter your ID : 1211153
this ID in array : 1 2 1 1 1 5 3
Do you have partners in your team ? Yes
Enter your ID:1212074
this ID in array:1 2 1 2 0 7 4
MENU
1. DISPLAY
2. COUNTS
3. SUM
4. MEAN
5. MEDIAN
6. MAX
7. MIN
8. NEW ID
9. EXIT
Menu option, choose wisely ...

```

Display, count, sum, mean , median, max, min, new id method body:

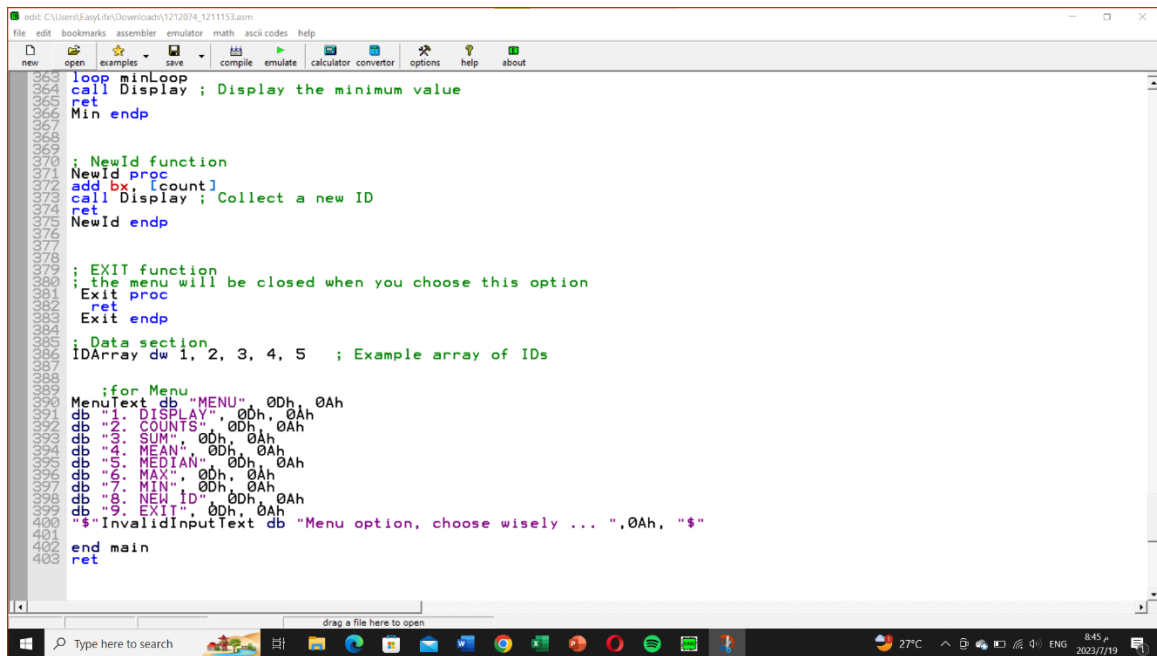
```
edit: C:\Users\ajay\Desktop\1212004_1211153.asm
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; Display function, display the IDs stored in the array.
Display proc
array db 1,2,1,2,0,7,4,1,2,1,1,1,5,3
mov ax,@data
mov ds,ax
mov si,offset array
mov cx,14
print "your value in array is : "
loopx:
mov di,[si] ; i arr[i] si [si]
add di,48
mov ah,02h
int 21h
mov di,32
mov ah,02h
int 21h
inc si
loop loopx
mov ah,04ch
int 21h
ret
Display endp

;Count function
Count proc
mov ax,[count] ; Move count to AX
call Display ; Display the count
ret
Count endp

; Sum function
```

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```
363 loop minLoop
364 call Display ; Display the minimum value
365 ret
366 Min endp
367
368
369
370 ; NewId function
371 NewId proc
372 add bx, [count]
373 call Display ; Collect a new ID
374 ret
375 NewId endp
376
377
378 ; EXIT function
379 ; the menu will be closed when you choose this option
380 Exit proc
381 ret
382 Exit endp
383
384 ; Data section
385 IDArray dw 1, 2, 3, 4, 5 ; Example array of IDs
386
387
388 ;for Menu
389 MenuText db "MENU", 0Dh, 0Ah
390 db "1. DISPLAY", 0Dh, 0Ah
391 db "2. COUNTS", 0Dh, 0Ah
392 db "3. SUM", 0Dh, 0Ah
393 db "4. MEAN", 0Dh, 0Ah
394 db "5. MEDIAN", 0Dh, 0Ah
395 db "6. MAX", 0Dh, 0Ah
396 db "7. MIN", 0Dh, 0Ah
397 db "8. NEW ID", 0Dh, 0Ah
398 db "9. EXIT", 0Dh, 0Ah
399 "$InvalidInputText db "Menu option, choose wisely ... ", 0Ah, "$"
400
401
402 end main
403 ret
```

Unfortunately the methods in the program are functioning but not producing the expected results, me and my partner re-viewed the algorithm of writing the code and cannot find where it goes wrong so it does not print the expected values.

VI. Conclusion

- As a first time writing an assembly code I could say me and my partner did a good job, although the methods weren't really working as it should, however I could say we developed knowledge into medium level programming using assembly language.
- The project implements a menu-driven program as described in the abstract it should take students ids and perform some operations on it according to the menu options.
- The work was 50/50 between me and my partner, all the code writing was on Ayah's laptop, but Rama also helped writing methods and sent it to Ayah.

First, we did the prompt from user “enter student id” then , asked if there is another partner this was Ayah’s work.

Rama provided the menu context.

The other methods were from both of us.

The report also both of us developed on it.

- Overall, this project was a good learning experience and helped us gain skills in assembly language, even though we still need to improve our writing but we are satisfied with the results and the project .